HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn] | **Stamp 1G** — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India **09/2021 – 01/2025 (3.5 Years)**

- Developed and validated **greenhouse gas emission models** for agricultural systems including **tillage, beef, dairy, pigs, poultry and mixed farming** using **Python** (pandas, NumPy, scikit-learn) and **SQL**-based data pipelines, achieving **98% accuracy** against IPCC methodologies across multiple geographies.
- Translated complex **agronomic and emissions science** into structured model logic and **clean quantitative modelling** frameworks integrated into production software platforms via **REST APIs** and database-level system integrations across **PostgreSQL, MySQL, and Oracle**.
- Collaborated with cross-functional software development teams to ensure **GHG models were accurately implemented, tested and maintained** as platform features evolved, reducing model validation time by **40%** through automated **ETL pipelines** and continuous integration via **Jenkins CI/CD**.
- Maintained current knowledge of **IPCC methodologies**, **national GHG inventory frameworks**, and **emerging carbon accounting standards** relevant to agriculture; documented methodology notes and technical specifications in **Confluence** for client-facing and certifying-body engagement.
- Supported expansion of modelling capabilities into new regions, commodities and supply chain use cases by designing **data warehousing** schemas, **Power BI dashboards** for emissions tracking, and **Tableau interactive visualizations** for stakeholder reporting, enabling **15+ agricultural domains** to measure Scope 3 emissions at field level.
- Leveraged advanced **statistical analysis**, **hypothesis testing**, **time-series modelling** and **machine learning** techniques (supervised/unsupervised learning, evaluation metrics) to validate emissions predictions, ensure scientific credibility, and support regulatory compliance across **SEPA**, **ISO 20022** and **data protection** frameworks in agricultural supply chains.

SKILLS

Greenhouse Gas Modelling & Carbon Science – IPCC methodologies, national GHG inventory frameworks, carbon accounting standards, agricultural emissions modelling (tillage, beef, dairy, pigs, poultry, mixed farming), Scope 3 emissions measurement, field-level carbon intelligence.

Data Engineering & Quantitative Modelling – Python (pandas, NumPy, scikit-learn), SQL (PostgreSQL, MySQL, Oracle), ETL pipeline development, data warehousing, statistical analysis, regression analysis, hypothesis testing, time-series modelling, machine learning (supervised/unsupervised), evaluation metrics.

Data Visualization & Business Intelligence – Power BI (dashboards, DAX measures, data modelling, Power Query), Tableau (interactive dashboards), exploratory data analysis, stakeholder reporting, analytics automation.

Software Development & Integration – Java Spring Boot, REST APIs, system integrations, database-level data flows, CI/CD (Jenkins, Git), Docker containerisation, AWS (EC2, S3, Lambda, CloudWatch), Linux shell scripting, Jira, Confluence.

Agricultural & Domain Knowledge – SEPA/instant payments/cross-border payments workflows, ISO 20022 message structures, agricultural supply chain systems, ERP integration support, change-control management, regulatory compliance (GDPR, data protection frameworks).

Documentation & Scientific Communication – methodology notes, technical specifications, user guides, training materials, cross-functional stakeholder engagement with agronomists, software developers, and certifying bodies.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin **02/2026**

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India **01/2021**

PROJECTS

Agricultural GHG Emissions Modelling Platform (HCL Technologies)

- Designed and deployed **greenhouse gas emission models** for multiple agricultural commodity types using **Python-based quantitative modelling** aligned with **IPCC methodologies** and **national GHG inventory frameworks**, enabling accurate measurement of farm-level and supply-chain emissions.
- Translated **complex agronomic science** into structured model logic integrated into production systems via **REST API endpoints** and **PostgreSQL/Oracle data layers**, supporting real-time emissions calculation across **15+ regions and commodity categories**.
- Built **Power BI dashboards** and **Tableau visualizations** to present emissions data to agricultural stakeholders and certifying bodies; automated data validation and model testing via **Python ETL pipelines** and **Jenkins CI/CD**, reducing model-maintenance overhead by **40%**.
- Documented **methodology notes**, **technical specifications**, and **regulatory compliance mappings** (SEPA, ISO 20022, data-protection frameworks) in **Confluence** for client training and audit support.

Ireland Jobs Tracker – AI-Driven Resume & Carbon Analytics Integration (Personal Project)

- Engineered end-to-end **Python automation workflow** combining **Claude API (Anthropic)** integration, structured-JSON **prompt engineering**, and **data pipeline orchestration** (Apify scrapers → JD ingestion → Claude-tailored content generation → reportlab PDF export) deployed via **GitHub Actions CI** and **Vercel**.
- Applied advanced **prompt engineering** and **AI workflow orchestration** techniques (constraint guardrails, temperature tuning, token budgeting) to generate job-description-tailored content at scale, demonstrating hands-on ability to translate domain-specific requirements (agricultural science, emissions models) into machine-readable logic.
- Demonstrated **quantitative data analysis** and **time-series modelling** skills by building job-market analytics dashboards using **Python (pandas, NumPy)** and **SQL querying** to track hiring trends; structured outputs for **Power BI** integration, showcasing ability to translate technical outputs into stakeholder-ready visualizations.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)